

Mehak K Sandhu

416-254-9545 | sandhu.mehak032@gmail.com | mehaks.pages.dev

EDUCATION

University of Guelph

Bachelor of Computing in Computer Science

Guelph, ON

Sep 2023 – Apr 2027

TECHNICAL SKILLS

Languages: Java, Python, C, SQL (PostgreSQL), JavaScript, R

Tools & Technologies: Git, Docker, Playwright, Snowflake, Salesforce

EXPERIENCE

Quality Engineer

Royal Bank of Canada (RBC)

May 2026 – Present

Toronto, ON

- Built a Snowflake-Salesforce data reconciliation tool in Python with configurable object mapping, multi-layer validators and HTML validation reports
- Automated cross-system discrepancy detection to replace manual QA checks, reduce validation efforts, and prevent release-blocking data sync issues with auditable output
- Self-initiated and shipped an AI-powered Confluence integrated tool that automated release-tracker reconciliation, reducing dev team release tracker maintenance time

Quality Engineer

Royal Bank of Canada (RBC)

May 2025 – Aug 2025

Toronto, ON

- Architected a Java/Spring Boot service virtualization layer that simulated RBC payment-engine APIs, enabling deterministic end-to-end QA without relying on live processing engines
- Replaced calls to real payment processors with mock services, eliminating accidental live transactions and improving test safety and execution speed

IT Technician

University of Guelph

Sep 2024 – Present

Guelph, ON

- Resolved hardware, network, and software issues for students and staff, applying systematic troubleshooting in a high-volume support environment

Automation Engineer

Prepr

Jun 2024 – Aug 2024

Toronto, ON

- Execute end-to-end testing across a full e-learning platform migration, identifying and documenting defects across functional areas including navigation, content delivery and user workflows
- Performed regression testing on bug fixes to verify resolution and prevent re-introduction, contributing to stable on-time feature releases

PROJECTS

Intelligent Python Tutor | Python, Next.js, PostgreSQL, scikit-learn

GitHub | Live

- Built an adaptive e-learning platform in Python that personalizes coding practice based on student performance using a modular ML layer
- Implemented a scikit-learn microservice for knowledge tracing and dynamic question difficulty adjusting, exposed via REST API
- Designed a rules-based tutoring engine as a fallback, keeping the ML layer decoupled and optional for graceful degradation

311 Complaint Patterns & Response Times | Python, Java, TypeScript, MySQL

GitHub | Live

- Built an interactive geospatial map dashboard (typescript frontend, Java backend) to visualize NTC 311 complaint trends, resolution times and service hotspots
- Developed a Python ingestion pipeline to fetch data from the NYC Open Data API and load into MySQL on a configurable refresh cycle
- Implemented multi-dimensional filtering to support exploratory data analytics